Question 3 — Examiner Q&A;

Anticipated questions and worked answers · SIMC 2026 Endeavour Challenge · May 2026

This document collects the questions a judging panel is most likely to ask about Question 3, each with a full written answer. It is organised by task (now covering 3a–3f) and is extended as further parts (3g, 3h) are completed. It accompanies the notebook *src/task_3.ipynb* and the technical guide *Task3_explained_simply.pdf*.

Task 3a — Cleaning the data

1. Why did you transpose the data file first?

When we load the file, its shape is 3401 by 1000. But the problem defines the design matrix with samples as the rows, and there are 1000 vials. So the axis of length 1000 must be the samples, which means the file was stored the other way round, with features along the rows. We therefore transpose it, swapping rows and columns, so each row becomes one vial and each column one frequency. Without this, every later operation along rows or columns would act on the wrong axis, and the whole analysis would silently be wrong.

2. The problem says 3041 features, but the data has 3401. Which is right?

The problem text says 3041 features, but the file actually contains 3401 columns. Those two numbers are the same digits in a different order, so this is almost certainly a simple typing slip in the question. We trust the data file rather than the prose, use 3401 throughout, and identify the samples by finding the axis of length 1000, which matches the stated number of vials. Letting the data itself settle such ambiguities, rather than the surrounding text, is the safe habit in any analysis.

3. What is the row norm, and why does it detect empty vials?

The norm of a spectrum is the square root of the sum of the squares of all its values, a single number describing the overall size, or energy, of that measurement. An empty vial holds no sample, so the instrument records only faint background noise, and its norm is tiny. A real drink produces tall absorption peaks, so its norm is large. Because the norm squares every value before adding them, the few large peaks dominate the sum, which makes the separation between empty and genuine vials extremely sharp and therefore easy and reliable to detect with a single threshold.

4. How did you choose the empty-vial threshold, and why not use a fixed value?

Rather than picking a fixed cut-off number, which we would then have to justify, we sort all 1000 norms and look for the largest empty gap between consecutive values. The empty vials sit at a norm of about 1.6, and then there is a wide gap before the real vials begin near 4.3. We place the threshold in the middle of that gap. This is robust, because a small change in the data does not move the boundary, and it removes any suggestion that we hand-tuned a magic constant simply to obtain the answer we were hoping to find.

5. How do you know the 40 vials are empty, and not just weak real samples?

Three signs confirm they are truly empty rather than merely faint real drinks. First, the gap is wide and perfectly clean: nothing at all sits between norm 1.7 and 4.3, so there is no smooth continuum of progressively weaker samples. Second, their spectra are flat noise, showing none of the sharp absorption peaks that a genuine drink produces. Third, about half of their readings are exactly zero, which is the fingerprint of zero-centred noise clipped at zero, whereas genuine spectra are only about a quarter zeros because their real peaks push the values well upward.

6. Why treat the 36 high-norm vials as an error? Could they be a real strong sample?

They might genuinely be a strong drink, and we say so openly. We flagged them as a likely error for two structural reasons: they are separated from the bulk by a gap about five times wider than any gap inside the data, and they form a solid block of rows immediately before the empty vials, suggesting both faulty groups were appended together. To stay honest, we showed they form a single tight, isolated cluster, we explicitly noted the alternative interpretation, and we made their removal reversible with one switch, so that the rest of the analysis is unaffected whichever reading is correct.

7. How did you validate the removals?

We used three independent checks. Visually, the removed spectra are flat noise while the kept spectra show clear peaks, so the two groups look qualitatively different. Statistically, the empty vials are about 50% exact zeros versus about 25% for real spectra, a quantitative signature of pure background noise. And structurally, when we later view the data in reduced form, removing the high-norm group deletes exactly one isolated cluster of points while leaving every other cluster completely untouched, which shows the removal is surgical and does not accidentally damage any of the genuine chemical structure we care about.

8. Why use a keep-mask instead of deleting rows?

A boolean mask is simply an array of true-or-false flags, one per vial, marking which to keep. We index the data with this mask instead of deleting rows, for three reasons. It is non-destructive, so the original matrix is never altered and any decision can be undone. It is composable, so two independent rules, not empty and not too loud, combine in a single clean expression. And it preserves the original row numbers, so a given vial keeps its identity through every later step, which matters whenever we want to trace a point in a plot back to one specific physical measurement.

Task 3b — Variance and the covariance matrix

9. What does mean-centering do, and why is it necessary?

Mean-centering subtracts each feature's average value, computed across all the kept vials, from every entry in that column, so that each feature then averages exactly zero. We do this because we care about how the drinks differ from one another, not about the large average spectrum they all share. If we skipped it, the single strongest pattern found later would simply be that shared average, which is identical for every vial and therefore tells us nothing about differences. Centering forces the analysis to describe deviations from the mean, and it is precisely in those deviations that all the discriminating information lives.

10. Why does the diagonal of the covariance matrix equal the variance?

The covariance matrix is defined as one over the number of samples, times the centred data transposed, times the centred data. A diagonal entry is the covariance of a feature with itself, which works out to one over the sample count times the sum of that feature's squared centred values. That expression is exactly the definition of variance. So the numbers running down the diagonal of the covariance matrix are precisely the per-feature variances, which is why, to answer the variance question, we need only that diagonal and never have to build or store the entire enormous square matrix.

11. Why divide by the sample count rather than the count minus one?

When computing a variance you can divide either by the number of samples, or by the number of samples minus one. Dividing by the full sample count gives what is called the population variance, and this is the convention the problem uses in its definition of the covariance matrix. Dividing by the count minus one gives the sample variance, used when estimating a wider population from a small subset. We deliberately divide by the full count, which is also NumPy's default, so that our variances exactly match the covariance definition and the later check, that their sum equals the trace, agrees.

12. What is the trace of the covariance matrix, and why check it?

The trace of a square matrix is the sum of the entries on its diagonal. For the covariance matrix, that sum is the total variance of the entire dataset. This total has a special property: rotating the coordinate axes redistributes variance among the new directions but can never create or destroy it, so the total is conserved. Because an eigen-decomposition is just such a rotation, the sum of all the eigenvalues must equal the trace. We verified that both equal about 7.9, which confirms that our decomposition was computed correctly and that no variance leaked away.

Task 3c — PCA and the eigenspectrum

13. Why use eigh instead of eig?

NumPy offers two routines for eigenvalues. The general one works on any square matrix but can return small imaginary parts caused by rounding, and it is slower. The specialised one is designed for symmetric matrices like our covariance matrix. It is guaranteed to return real eigenvalues and a set of perfectly perpendicular eigenvectors, and it is faster and numerically more stable because it exploits the symmetry. Since the covariance matrix equals its own transpose, it is symmetric, so the specialised symmetric solver is both the correct and the safer choice, and it is the one the problem hint recommends.

14. In plain terms, what is an eigenvalue here?

An eigenvalue, in this setting, is the amount of variance the data has along the direction of its eigenvector. Each eigenvector is one candidate pattern, or axis, drawn through the cloud of spectra, and the matching eigenvalue measures how spread out the data is along that axis. A large eigenvalue means that direction captures a great deal of the variation between drinks and is therefore important. A tiny eigenvalue means almost nothing varies along that direction, so it represents noise. Ranking the eigenvalues from largest to smallest therefore ranks the underlying patterns from the most informative down to the least.

15. Why clip negative eigenvalues to zero?

The covariance matrix is positive semi-definite, which means that mathematically all of its eigenvalues must be greater than or equal to zero, since each one represents a variance and a variance can never be negative. However, when a computer calculates them in finite precision, rounding error can produce values like -10^{-17}, negative only by a vanishing amount. We clip these up to zero so the values remain physically meaningful. This also lets us safely take their square roots later, when those eigenvalues are reinterpreted as the squared singular values from Question 2.

16. How did you pick the elbow at eight, and why not seven or nine?

We locate the elbow as the largest multiplicative drop between consecutive eigenvalues, and that biggest drop, a factor of about four, falls between the eighth and the ninth. So the elbow is at eight. It is not seven, because the eighth eigenvalue still sits about four times above the flat noise floor, marking it clearly as real signal that we ought to keep. And it is not nine, because the ninth eigenvalue already lies down on the noise floor, statistically indistinguishable from the hundreds of tiny noise eigenvalues that follow it. Eight is the point that cleanly separates signal from noise.

17. What does a first-to-second eigenvalue ratio of about 3.5 tell you?

The ratio of the first eigenvalue to the second is about 3.5, meaning the single most important pattern captures 3.5 times as much variance as the next one. A large ratio tells us that one direction strongly dominates the variation, so the data has a powerful principal axis and is effectively low-dimensional. The same quantity, viewed as the spectral gap, also governs the Power Method from Question 2: a wide gap makes that iterative algorithm converge quickly and robustly, which links this single observation directly back to the earlier theoretical part of the competition.

18. Why is the keep 95% of the variance rule misleading here?

A common rule of thumb keeps as many components as are needed to explain 95% of the total variance. Here that rule is misleading. The eight genuine signal components explain only about 55%, and to reach 95% you would need 667 components. The reason is that although each individual noise eigenvalue is tiny, there are roughly ~900 of them, and their tiny contributions add up to a large share of the total. So the 95% rule would wildly overestimate the true dimensionality, whereas the elbow at eight remains the honest, defensible measure.

Task 3d — The leading eigenvector

19. Why decompose the large covariance matrix instead of the smaller Gram matrix?

The covariance matrix and the Gram matrix share the same non-zero eigenvalues, and the Gram matrix is smaller, so for the eigenvalues alone it would be cheaper. However, we specifically need the eigenvectors that live in feature space, the directions among the frequencies, because those are exactly what we interpret in part d and project the samples onto in part e. Decomposing the covariance matrix hands us those feature-space eigenvectors directly. The cost is about 14 seconds, paid only once, and the result is reused throughout the remaining parts, so the modest extra effort is clearly worthwhile.

20. What does the leading eigenvector represent?

The leading eigenvector is the single direction in frequency space along which the drinks differ from one another the most. It assigns a weight to every one of the 3401 frequencies, describing the most important pattern of variation. Its practical value is that if you had to summarise an entire spectrum with just one number, the best possible choice is the projection of that spectrum onto this direction. Among all possible directions, it is the one whose resulting summary numbers are the most spread out, and therefore the most informative single descriptor you can attach to a vial.

21. Why does the leading eigenvector have negative components?

The leading eigenvector contains both positive and negative weights, with about a quarter of them negative. A direction with mixed signs represents a contrast rather than an overall level. Concretely, vials that score high in the positively weighted frequency bands tend to score low in the negatively weighted bands, and the other way around. So this pattern does not measure how strong or bright a drink is overall; it measures the balance of absorption between two groups of frequencies. The sign of a vial's score then tells you which side of that underlying chemical contrast the particular drink falls on.

22. Why use cosine similarity instead of Euclidean distance to compare it with spectra?

We wanted to know whether the leading eigenvector has the same shape as some real measurement, and shape is about direction, not size. Cosine similarity is the cosine of the angle between two vectors: it is one when they point the same way, zero when they are perpendicular, and minus one when opposite, and crucially it ignores their lengths entirely. That is exactly right here, because the eigenvector has length one while the spectra do not. Euclidean distance, by contrast, would be dominated by the large difference in magnitude between them and would end up answering the wrong question.

23. Why does it match a centered spectrum, but not a raw one or the average?

The leading eigenvector matches the centred spectrum of vial 193 with a cosine of 0.92, but only 0.67 against that vial's raw spectrum, and almost zero against the average spectrum. The explanation is that the eigenvector lives in the mean-centred space and is therefore essentially perpendicular to the average spectrum. Raw spectra are dominated by that shared average, so the eigenvector cannot resemble them. Instead it resembles a difference-spectrum: the way the most unusual drink departs from the average, which is precisely what mean-centering was designed to expose.

Task 3e — Clustering in covariance space

24. Why project onto exactly four principal components?

We keep the four leading eigenvectors and give each vial four coordinates, namely its scores on those four patterns. Four is a deliberate choice. It lies comfortably within the eight genuine signal directions identified by the elbow, so we retain the real structure and discard the noise. Yet four is small enough to visualise completely, because four coordinates produce only six distinct pairwise plots, which we can lay out and inspect by eye. More components would add very little extra signal while making the visual inspection unwieldy, and the problem itself specifically asks for the top four components.

25. In the scatter plots, what does one point represent, and why small transparent markers?

In each scatter plot, one point represents one vial, positioned according to its scores on the two principal components shown on the axes. We draw the points small and partly transparent. Transparency matters because many vials of the same recipe land almost exactly on top of one another; with solid markers they would merge into a single opaque blob and we could not tell a dense cluster from a sparse one. With semi-transparent markers, overlapping points deepen the colour, so dense clusters appear as darker regions and the true grouping structure of the data stays clearly visible.

26. How did you count the clusters without a clustering library, and is it robust?

Because the environment had no clustering library installed, we wrote a simple method called leader clustering. We walk through the points one at a time, and whenever a point lies farther than a chosen radius from every existing cluster centre, we start a new cluster there. The only parameter is that radius. We set it to a few times the typical distance between nearest neighbours, and reassuringly the resulting count stays fixed at 25 for any radius from six to twelve times that spacing. Because the answer is stable across such a wide range, it reflects the data, not the threshold.

27. How can feature-space eigenvectors reveal groups among the samples?

It seems surprising, because the eigenvectors are directions in frequency space, yet they reveal groups among the samples. The bridge is projection. Each sample's scores are the dot products of its spectrum with those eigenvectors, and these scores serve as the sample's coordinates in a small principal subspace. Two vials with nearly identical spectra produce nearly identical scores, so they land close together in that subspace. Therefore clusters of points in the projected picture correspond to groups of chemically similar vials, even though the axes themselves are constructed entirely from the frequencies rather than from the samples.

Task 3f — Clustering via the Gram matrix

28. What is the Gram matrix, and how does it differ from the covariance matrix?

The Gram matrix is $G = (1/\tilde{N})\ \tilde{X}\ \tilde{X}^{T}$, an $\tilde{N}\times\tilde{N}$ table (924×924) that lives in sample space; its entry G_ij is the dot-product similarity between vials i and j. The covariance matrix $C = (1/\tilde{N})\, \tilde{X}^T\tilde{X}$ is M×M and lives in feature space, telling you which frequencies co-vary. So G compares samples to samples while C compares features to features. They are built from the same centred data in opposite orders and, crucially, share the same nonzero eigenvalues, so they encode one underlying structure seen from two complementary viewpoints.

29. Why can you read the sample coordinates straight off the Gram eigenvectors, with no projection?

Because the Gram matrix already lives in sample space: each of its eigenvectors is an $\tilde{N}$-dimensional vector with one component per vial. Component i of the k-th Gram eigenvector is therefore directly the coordinate of vial i along principal axis k — there is nothing left to project. This contrasts with the covariance matrix, whose eigenvectors are M-dimensional directions in feature space; there you must form the dot product $\tilde{x}_i \cdot \nu_k$ (a projection) to obtain each sample's coordinate. The Gram route hands you the sample scores immediately.

30. How are the Gram and covariance eigenvectors related, and why do they give the same clusters?

Through the singular value decomposition $\tilde{X} = U\Sigma V^T$. The covariance eigenvectors are the right singular vectors V (feature space); the Gram eigenvectors are the left singular vectors U (sample space); both have eigenvalues $\sigma^2/\tilde{N}$, hence identical nonzero eigenvalues. The covariance-space scores $\tilde{X}V$ equal $U\Sigma$, while the scaled Gram eigenvectors $U\sqrt{(\tilde{N}\lambda)}$ reproduce the very same coordinates. So the two scatter plots are the same up to a per-axis sign (eigenvectors are defined only up to sign), which is exactly why both methods resolve the same 25 clusters.

31. You scaled the Gram eigenvectors by $\sqrt{(\tilde{N}\lambda)}$. Why, and does it change the cluster count?

A raw Gram eigenvector is unit-length, so its components are tiny ($\approx 1/\sqrt{924}$). Scaling each by $\sqrt{(\tilde{N}\lambda_k)}$ restores the natural spread of the data along that axis and makes the coordinates numerically equal to the covariance-space scores $\tilde{X}V$ of 3e, so the two scatter plots are directly comparable. It does not change the cluster count: our leader-clustering radius is set as a multiple of the median nearest-neighbour spacing, so it rescales with the data. The count stays 25 across radii from six to twelve times that spacing.

32. Why might the Gram-space scatter look flipped compared with 3e?

Eigenvectors are defined only up to an overall sign: if u is a unit eigenvector then so is −u, with the same eigenvalue. The symmetric solver may return either, independently for each axis, so a Gram principal component can come out as the negative of the matching covariance one — which flips that axis of the scatter. Where eigenvalues are nearly tied the eigenvectors can also rotate among themselves. None of this changes the distances between points, so the clusters and their count are unaffected; only the orientation of the picture differs.

33. The covariance matrix is 3401×3401 and the Gram matrix is 924×924 — why compute both?

They share the same nonzero eigenvalues, so for the eigenvalues alone the Gram matrix is far cheaper (924×924 versus 3401×3401). We still build the covariance matrix because its eigenvectors live in feature space, which we need in 3d to interpret which frequencies drive ν_1 and in 3e to project. The Gram matrix is the natural choice when only sample-space structure (the clustering) is wanted, because its eigenvectors are the sample coordinates directly. Choosing between them is a question of which space you need and which is cheaper to diagonalise.

General and cross-question links

34. How does Task 3 connect to Questions 1 and 2?

Question 1 studied a Markov process whose dominant eigenvector was the long-term steady state, the direction the system settles into over time. Here the dominant eigenvector plays the analogous starring role: it is the main axis along which the drinks differ. Question 2 introduced the Power Method, an iterative way to find that dominant eigenvector without a full decomposition, and showed that it converges at a rate set by the spectral gap. Our covariance matrix has a wide spectral gap, the ratio of about 3.5, so the same Power Method would converge here quickly and stably.

35. What is the difference between the covariance and Gram matrices?

Both matrices are built from the same centred data, but in opposite orders. The covariance matrix is the centred data transposed times itself, giving a feature-by-feature table that lives in frequency space and tells you which frequencies vary together. The Gram matrix is the centred data times its own transpose, giving a sample-by-sample table that lives in vial space and tells you which measurements resemble one another. They are two views of the same underlying structure and share the same non-zero eigenvalues, so choosing between them is mainly a question of computational cost and numerical convenience.

36. Could the cleaning have removed a real cluster by accident?

For the empty vials there is no real risk, because they are structureless flat noise containing no chemical information, so removing them cannot discard a genuine group. For the high-norm vials the concern is fair, and we took it seriously. We demonstrated that they form one single, tight, isolated cluster rather than scattered points, that removing them lowers the cluster count by exactly one while every other cluster stays intact, and we kept the removal reversible. So even if they turned out to be a real strong drink, our main conclusions about all the other groups would be unaffected.